Generative Models as Recommendation Engines: A Survey of Architectures, Representations, and Learning Paradigms

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Abstract

Generative recommendation has emerged as a promising approach that uses generative models to predict user preferences and generate recommendations. These models offer new ways to represent items and model user behavior beyond conventional candidate scoring and ranking. However, existing methods differ considerably in their objectives, output representations, and inference strategies, complicating comparisons of their relationships and practical trade-offs. In this survey, we review three major research directions: language-model-based recommendation, recommendation-specific generative systems, and diffusion-based recommendation. We compare representative methods in terms of item representation, learning objectives, generation strategies, and computational efficiency. We also discuss challenges in valid item generation, dynamic catalog maintenance, inference cost, and reliable evaluation. Through this comparison, we examine each paradigm's strengths, limitations, and integration into practical recommender systems. Our analysis clarifies recent advances in model-level generative recommendation and identifies directions for future research.

1 Introduction

Recommender systems play an important role in helping users discover relevant items from large and continuously evolving catalogs. Traditional recommender systems typically rely on a multi-stage pipeline consisting of candidate retrieval, ranking, and filtering. For example, the YouTube recommender system separates candidate generation from ranking to efficiently handle large-scale recommendations [3]. Although this architecture has proven effective, independently optimized stages may introduce inconsistencies between retrieval and ranking objectives. In particular, potentially relevant items can be excluded during early stages and cannot be recovered by subsequent ranking. Generative recommendation offers an alternative approach by using generative models to learn user preferences and generate recommendations. Instead of relying solely on scoring candidate items, these methods can generate item identifiers, recommendation sequences, or preference representations that are subsequently mapped to items in the catalog. This shift introduces new opportunities for recommendation modeling while also raising challenges in representation learning, inference, and system efficiency.

Recent advances in generative modeling have led to an increasing number of recommendation approaches. Language-model-based methods formulate recommendation as a language generation task, allowing models to use semantic information and pretrained knowledge. Recommendation-specific generative systems explore item tokenization, generative retrieval, and large-scale sequential modeling to produce recommendation outputs. Diffusion-based approaches provide another perspective by learning user preferences or item representations through iterative denoising. Although these methods share the goal of improving recommendation through generative modeling, they differ considerably in their output representations, learning objectives, and inference procedures. Existing surveys provide broad overviews of generative recommendation [4, 43] or focus specifically on language-model-based approaches [17]. However, a focused comparison of how different generative paradigms generate recommendations and function within practical recommendation pipelines remains valuable.

In this survey, we focus on *model-level generative recommendation*, where generative models play a central role in producing recommendation decisions. This scope excludes methods used solely for data augmentation, feature extraction, or explanation generation. We review three major research directions: language-model-based recommendation, recommendation-specific generative systems, and diffusion-based recommendation. Representative approaches include P5 [6], TIGER [33], HSTU [41], OneRec [5], DiffRec [37], and DreamRec [40], together with related studies on item tokenization, catalog grounding, and generative learning. We compare these approaches in terms of item representation, learning objectives, generation strategies, and computational efficiency. We also examine challenges related to valid item generation, dynamic item catalogs, inference costs, and reliable evaluation. To account for differences in experimental settings, we focus on the underlying technical distinctions and trade-offs among these methods.

Figure 1 summarizes the organization of this survey. The three major research directions are organized into the subtopics examined in Section 3, with representative methods shown for each subtopic. We then compare these approaches in Section 4 and discuss open challenges in Section 5.

2 Background and Formulation

Let $\mathcal{I}$ denote the item catalog, $h = (i_1, a_1, \ldots, i_t, a_t)$ a user's interaction history, where i_t and a_t represent the interacted item and the corresponding action, and c the recommendation context. Conventional recommendation methods typically learn a scoring function $s_\theta(h, c, i)$ to estimate the relevance of a candidate item $i \in \mathcal{I}$ to user preferences, and produce recommendations through candidate retrieval and ranking [3].

Generative recommenders model user preferences using $p_\theta(y \mid h, c)$, a conditional distribution over recommendation-related outputs, where y may represent a natural-language item description, a discrete item identifier, a sequence of semantic tokens, or a continuous preference representation. For a discrete token sequence

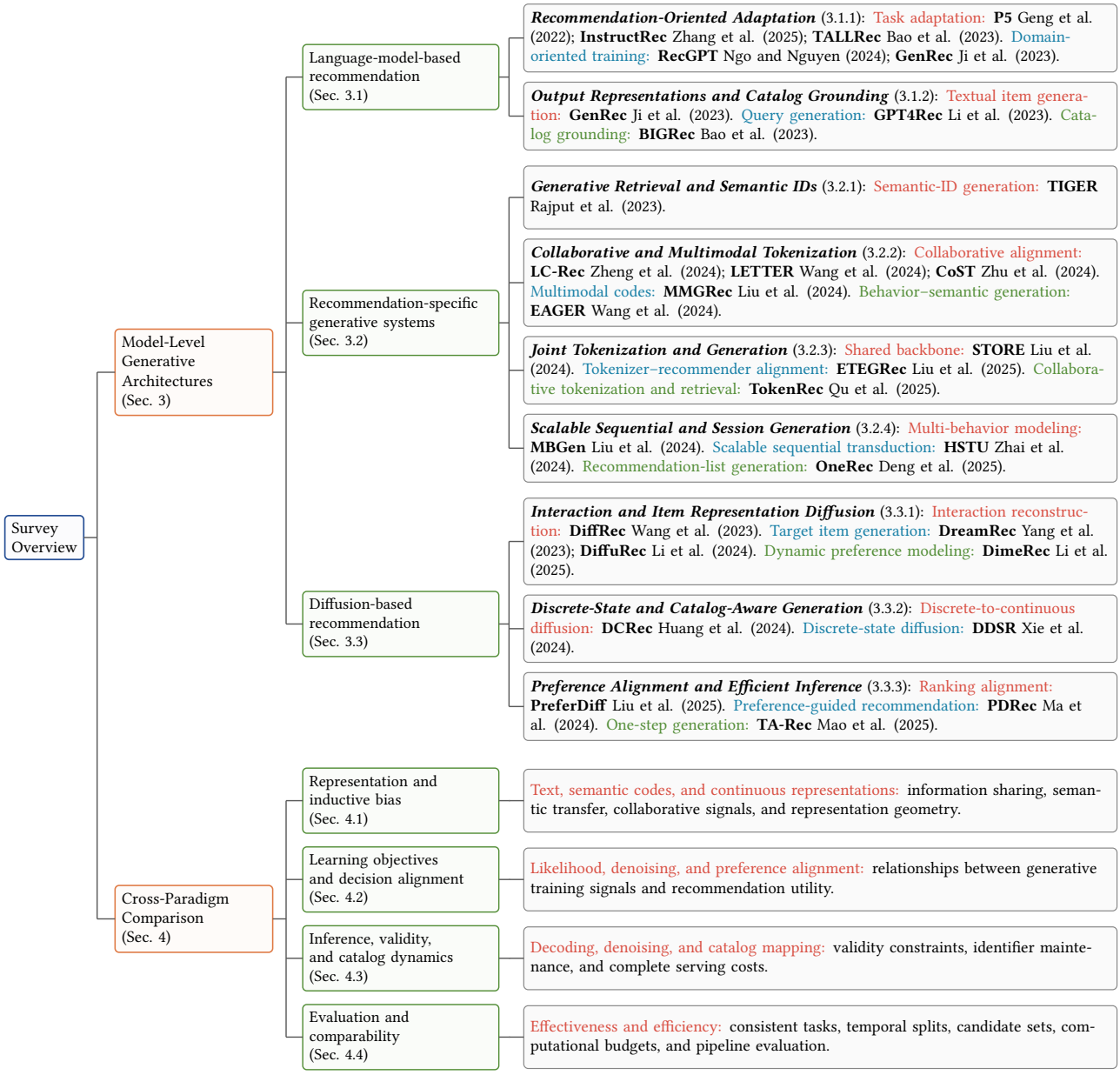


Figure 1: Taxonomy and organization of this survey. The tree follows the sections on generative architectures and cross-paradigm comparison. Representative methods are grouped by the architectural subtopics reviewed in the text; colored labels identify their technical focus. The grouping follows the survey structure, and related methods may span multiple topics.

$y = (y_1, \ldots, y_L)$, autoregressive models factorize the conditional distribution as follows:

$$p_\theta(y \mid h, c) = \prod_{j=1}^{L} p_\theta(y_j \mid y_{<j}, h, c). \tag{1}$$

Autoregressive approaches typically maximize the likelihood of target sequences, whereas diffusion-based approaches learn continuous user preference or item representations through iterative denoising. Different generative mechanisms lead to different recommendation procedures. Textual outputs and discrete identifiers can be resolved to catalog items through decoding and lookup, while continuous representations are typically converted into recommendations through similarity-based retrieval or preference-score ranking. Figure 2 places these procedures side by side: each paradigm produces a different intermediate output, and a catalog-mapping step turns that output into the final top-K items.

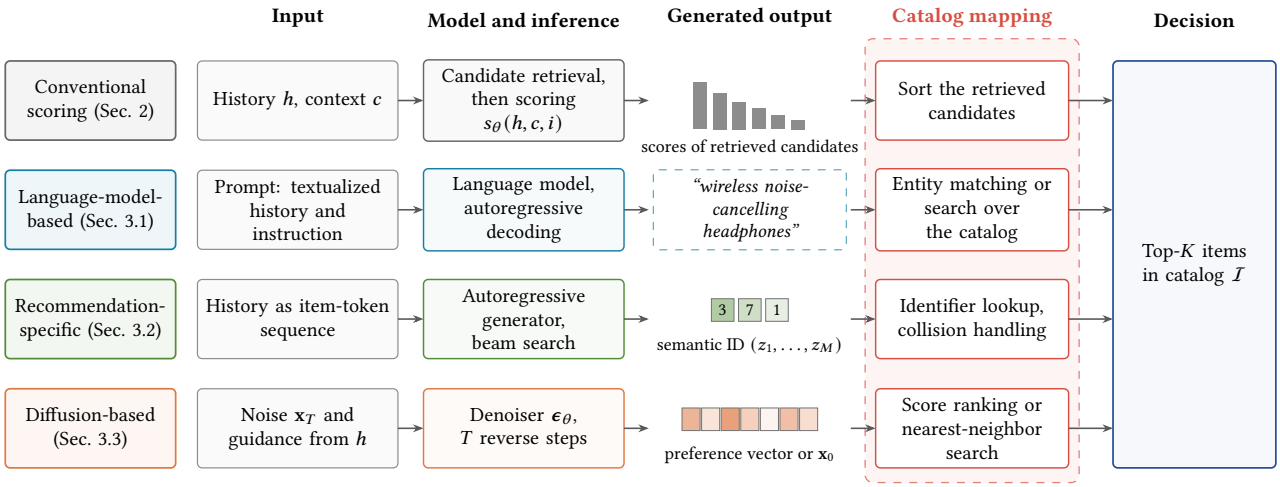


Figure 2: Recommendation procedures of conventional scoring and the three generative paradigms. The paradigms differ in what the model generates and in how the generated output is mapped back to items in the catalog (shaded column).

Generative modeling has a long history in recommender systems. Mult-VAE [20] employs a variational autoencoder to probabilistically model user interactions and rank items using reconstructed preference signals. SASRec [13] and BERT4Rec [34] learn sequential user preferences through causal self-attention and bidirectional contextual modeling, respectively, providing important foundations for subsequent research on generative sequential recommendation. With advances in language models, semantic identifiers, and diffusion models, the generation targets of recommender systems have expanded to item tokens, recommendation sequences, and continuous latent representations, giving rise to diverse generative modeling paradigms.

This survey focuses on *model-level generative recommendation*, in which generative models directly contribute to recommendation decisions. We examine three major paradigms: language-model-based recommendation, recommendation-specific generative systems, and diffusion-based recommendation. The following sections analyze these paradigms along four dimensions: output representation, learning objective, inference mechanism, and recommendation decision process, highlighting their modeling characteristics and practical applicability.

3 Model-Level Generative Architectures

This section reviews the three paradigms in turn.

3.1 Language-Model-Based Recommendation

Language-model-based recommendation formulates recommendation tasks as conditional language generation, enabling user interactions, item attributes, and task instructions to be represented within a unified textual framework. This paradigm leverages pretrained linguistic knowledge to capture semantic relationships among items while learning personalized preferences from interaction data. Existing approaches primarily differ in how language models are adapted to recommendation tasks and how generated outputs are mapped to actual catalog items.

3.1.1 Recommendation-Oriented Adaptation. P5 [6] establishes a unified text-to-text paradigm that represents diverse recommendation tasks through personalized prompts and trains a shared sequence-to-sequence model. By expressing user interactions, item metadata, and prediction targets in natural language, P5 enables different tasks to share representations and learning objectives. This formulation provides a flexible interface for recommendation, although the quality of personalized predictions remains dependent on the interaction signals available during training.

Subsequent research investigates how pretrained language models can be adapted to recommendation-specific objectives. InstructRec [42] formulates recommendation as instruction following, incorporating personalized requirements into task descriptions to support diverse recommendation scenarios. TALLRec [2] adopts parameter-efficient fine-tuning to align language models with recommendation tasks using interaction data. These methods emphasize task adaptation through instruction supervision, allowing general language representations to support preference-oriented prediction.

Another direction incorporates recommendation knowledge more directly into model training. RecGPT [30] investigates domain-adapted generative pretraining and instruction tuning, strengthening the connection between language modeling and domain-specific data. GenRec [10] fine-tunes a LLaMA-based model on textualized user–item interactions and generates target items from historical preferences. Unlike approaches that primarily select among explicitly provided candidates, GenRec treats item prediction as a language generation task.

These studies reveal two complementary strategies for recommendation adaptation. Instruction-based methods emphasize the correspondence between task descriptions and expected outputs, whereas domain-oriented training incorporates interaction patterns

into the model's learned representations. Their effectiveness depends on the integration of linguistic semantics and collaborative preference signals. In particular, semantic relationships derived from item descriptions provide useful content information, while interaction histories capture user-specific preferences that may not be fully expressed through textual similarity.

3.1.2 Output Representations and Catalog Grounding. The output representation determines how language generation is translated into recommendation decisions. Language-model-based methods employ several forms of generated outputs, including item titles, descriptive text, and search queries. These representations provide different interfaces between the language model and the underlying item catalog.

GenRec [10] uses textual item identities as generation targets, allowing the model to associate recommendation outputs with natural-language item information. GPT4Rec [16] adopts a different formulation by generating search queries from user interaction histories and retrieving corresponding items through a search engine. Its multi-query generation strategy represents different aspects of user interests, connecting generative preference modeling with conventional information retrieval. The quality of the resulting recommendations therefore reflects both the generated queries and the retrieval mechanism.

BIGRec [1] explicitly separates recommendation generation from catalog grounding. It first adapts a language model to generate recommendation-oriented item descriptions and subsequently identifies corresponding catalog items. This two-stage formulation connects the semantic flexibility of language generation with the requirement that recommended items belong to an existing catalog. It also highlights the importance of the mapping mechanism in determining final recommendation quality.

Different output interfaces introduce distinct representation and inference characteristics. Natural-language descriptions can exploit semantic knowledge acquired during pretraining, but their correspondence to catalog entries may be ambiguous. Item titles may be shared by multiple products, while tokenization can produce variable-length representations. Catalog grounding resolves generated text into identifiable items through matching, retrieval, or ranking, introducing additional decisions into the recommendation pipeline. Consequently, the effectiveness of language-based recommendation depends on both the quality of preference generation and the accuracy of the grounding procedure.

Output representation also influences the scope of evaluation. Candidate-based recommendation measures selection within a specified item set, whereas full-catalog generation requires the model and its grounding mechanism to identify relevant items from the entire available catalog. Evaluating these settings separately helps distinguish task adaptation from retrieval capability. Existing benchmarks [24] further motivate assessing direct item selection alongside other language-oriented recommendation tasks.

Overall, language-model-based recommendation offers a flexible framework for combining semantic knowledge, personalized interaction histories, and task-specific instructions. Its central design choices concern the adaptation of linguistic representations to collaborative preferences and the connection between generated language and catalog-level decisions. These considerations also motivate recommendation-specific generative systems that replace natural-language outputs with structured item representations, as discussed in the following section.

3.2 Recommendation-Specific Generative Systems

Recommendation-specific generative systems represent items and user interactions using structured tokens or learned representations tailored to recommendation tasks. Unlike language-oriented approaches that primarily rely on textual outputs, these systems directly model item identities, interaction sequences, or recommendation lists within specialized output spaces. Their development has centered on four closely related directions: semantic item tokenization, collaborative representation learning, joint optimization of tokenization and generation, and scalable sequential generation.

3.2.1 Generative Retrieval and Semantic IDs. Generative retrieval reformulates candidate retrieval as the prediction of item identifiers conditioned on user interaction histories. TIGER [33] establishes this paradigm by representing each item as a sequence of discrete semantic tokens derived from its content embedding. A sequence-to-sequence Transformer then generates the semantic identifier of a relevant item autoregressively, replacing independent candidate scoring with token-level prediction.

The construction of semantic identifiers typically relies on residual vector quantization. Given an item embedding $\mathbf{e}_i$, a sequence of codebook vectors is selected to approximate its representation:

$$\mathbf{e}_i \approx \sum_{m=1}^{M} \mathbf{c}_{m,z_{i,m}}, \tag{2}$$

where M denotes the number of quantization levels, $z_{i,m}$ is the code index assigned to item i at level m, and $\mathbf{c}_{m,z_{i,m}}$ is its corresponding codebook vector. Each level quantizes the residual left by preceding levels. RQ-VAE [15] learns such hierarchical representations through reconstruction and quantization objectives.

The resulting identifier $(z_{i,1}, \ldots, z_{i,M})$ provides a compact representation of item identity. Items with related representations may share code prefixes, enabling statistical sharing across items during autoregressive prediction. At inference, beam search generates candidate code sequences, which are resolved into catalog items through identifier lookup. Additional disambiguation is used when multiple items receive the same semantic codes. Figure 3 illustrates both steps.

This formulation connects representation learning with generative retrieval: the tokenizer determines the structure of the prediction space, while the generator learns the conditional distribution over item codes. Consequently, retrieval effectiveness depends on both the semantic structure of the identifiers and the probability assigned to their token sequences. Codebook capacity, identifier collisions, and decoding strategies jointly affect retrieval accuracy and computational efficiency.

3.2.2 Collaborative and Multimodal Tokenization. Early semantic tokenization primarily emphasizes content-derived item representations. Subsequent research incorporates collaborative signals to better align item identifiers with personalized recommendation objectives.

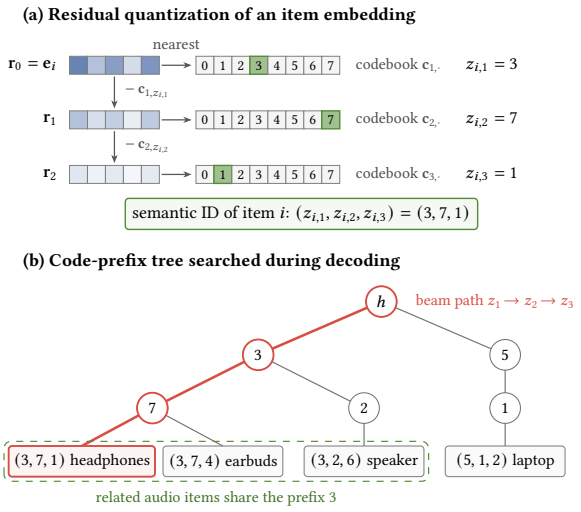


Figure 3: Semantic identifiers. (a) Each level selects the nearest codebook vector and passes the residual to the next level. (b) Related items share code prefixes, and decoding follows a beam path through the prefix tree to a valid item.

LC-Rec [44] addresses the gap between linguistic semantics and collaborative preferences by learning discrete item indices through vector quantization and introducing alignment tasks that connect item codes, language representations, and user interactions. The resulting identifiers provide a structured interface through which language models can incorporate collaborative information into item generation.

LETTER [36] further investigates the representation properties required for effective generative recommendation. Its tokenization framework combines hierarchical semantic regularization, collaborative alignment, and code-assignment diversity. The semantic objective preserves relationships among item representations, collaborative supervision incorporates interaction patterns, and diversity regularization encourages more balanced code usage. LETTER also introduces a ranking-guided generation objective to connect token prediction with recommendation relevance.

Related approaches explore alternative mechanisms for integrating collaborative structure. CoST [45] incorporates contrastive item relationships into the quantization process, providing additional supervision beyond embedding reconstruction. MMGRec [22] constructs item identifiers through graph-aware residual quantization of multimodal and collaborative representations, and subsequently generates these identifiers autoregressively. These methods extend semantic tokenization beyond content similarity toward preference-aware item representations.

The integration of heterogeneous information can also occur at the generation stage. EAGER [38] employs a shared encoder with separate behavioral and semantic decoders. The two streams generate complementary item representations, which are combined through confidence-based ranking. Whereas LETTER, CoST, and MMGRec emphasize the construction of informative item codes, EAGER incorporates behavioral and semantic information through parallel generative pathways.

Item indexing itself introduces another design dimension. Comparative experiments using the P5 backbone [8] demonstrate that different identifier schemes, including random, sequential, collaborative, and semantic representations, can substantially influence recommendation performance. IDGenRec [35] explores concise textual identifiers that preserve linguistic meaning while representing item identity. These indexing strategies connect identifier structure with the information available to the generative model.

Taken together, these studies establish item tokenization as an important component of generative recommendation whose effects extend beyond preprocessing. Effective identifiers must balance semantic information, collaborative relevance, code distinguishability, and generation efficiency. Empirical analyses such as GRID [12] further highlight the importance of implementation choices and component-level evaluation in semantic-ID-based recommendation.

3.2.3 Joint Tokenization and Generation. Conventional semantic-ID pipelines often optimize item tokenization and recommendation generation in separate stages. The tokenizer learns a discrete representation from item features, after which a generative model is trained to predict the assigned codes. This separation simplifies training but introduces a potential mismatch between representation fidelity and predictive utility: codes that reconstruct item embeddings accurately may not be optimal for prediction from user histories.

Recent approaches address this mismatch through closer integration of representation learning and recommendation objectives. STORE [25] formulates semantic tokenization and generative recommendation within a shared language-model backbone. It unifies text-to-token and token-to-token generation with auxiliary reconstruction tasks, allowing item representation learning and recommendation prediction to share model parameters.

ETEGRec [21] couples an item tokenizer with a generative recommender through two complementary alignment objectives. Sequence–item alignment connects generated sequence representations with target items, while preference–semantic alignment relates collaborative preferences to item semantics. Alternating optimization coordinates the two components, enabling token representations to evolve with the recommendation objective.

TokenRec [31] approaches the problem through collaborative tokenization and efficient retrieval. It quantizes masked user and item representations learned from collaborative information, while introducing a retrieval mechanism that avoids conventional autoregressive beam decoding. Its design connects the quality of discrete representations with the computational requirements of top-K recommendation.

These methods represent complementary directions for improving the relationship between tokenization and generation. STORE emphasizes architectural sharing, ETEGRec optimizes the interaction between separate components, and TokenRec jointly considers collaborative token construction and retrieval efficiency. Their differences indicate that representation quality should be assessed through downstream recommendation performance and decoding cost, in addition to embedding reconstruction.

Another consideration is the stability of item identifiers as catalogs evolve. Updating learned codebooks may change the token sequences associated with existing items, affecting previously learned generation patterns. Maintaining stable identifiers, incorporating newly introduced items, and updating catalog mappings therefore form part of the representation-learning problem. These requirements connect tokenizer design with the long-term operation of generative recommendation systems.

3.2.4 Scalable Sequential and Session Generation. Beyond item representation, recent generative systems investigate how recommendation objectives and model architectures can support heterogeneous behaviors, long interaction histories, and list-level prediction.

MBGen [27] extends autoregressive recommendation to multi-behavior interaction sequences. It interleaves behavior and item tokens and models recommendation as a two-stage prediction process: generating the next behavior type and subsequently predicting an item conditioned on that behavior. A position-routed sparse architecture supports efficient modeling of the heterogeneous token sequence. This formulation incorporates behavioral intent directly into the generative prediction objective.

HSTU [41] investigates the scalability of sequential recommendation through a generative transduction framework. Its hierarchical sequential transduction units are designed for high-cardinality, nonstationary interaction data, with architectural and serving optimizations supporting long histories and large-scale computation. Unlike semantic-ID retrieval models, HSTU primarily advances the modeling and computational efficiency of user action sequences. It illustrates how generative sequential modeling can be integrated into industrial recommendation architectures without requiring the same identifier-generation interface as TIGER.

OneRec [5] extends generative recommendation from candidate retrieval toward unified recommendation-list generation. Its encoder–decoder architecture generates sequences of recommended items, while a sparse mixture-of-experts design increases model capacity. The system further introduces iterative preference alignment to optimize generated sessions according to learned preference signals.

The distinction between next-item and session-level generation can be expressed through the probability of an ordered recommendation list $Y = (i_1, \ldots, i_K)$:

$$p_\theta(Y \mid h) = \prod_{k=1}^{K} p_\theta(i_k \mid h, i_{<k}). \tag{3}$$

Here, each generated item is conditioned on the interaction history and preceding recommendations. This factorization allows list generation to model dependencies among recommended items, including complementarity, substitution, and redundancy. It therefore extends the prediction target from individual item relevance to the structure of an entire recommendation sequence.

OneRec [5] further incorporates iterative preference alignment based on direct preference optimization [32]. A reward model evaluates generated sessions and provides preference comparisons for optimizing the relative likelihood of preferred and rejected sequences. This mechanism supplements sequence-likelihood training with feedback associated with session-level recommendation quality.

Scalability and integration take several forms in these systems. MBGen expands the behavioral information represented in generative sequences, HSTU addresses the computational demands of large-scale interaction modeling, and OneRec extends generation toward list-level recommendation and preference alignment. Together, they demonstrate the progression from item-oriented generative retrieval toward more expressive recommendation objectives and integrated serving architectures.

Overall, recommendation-specific generative systems advance the field along two complementary axes: learning structured representations that support accurate item generation, and developing generative architectures that capture increasingly complex recommendation decisions. Their practical effectiveness depends on the interaction among identifier design, learning objectives, decoding strategies, and system-level efficiency.

3.3 Diffusion-Based Recommendation

Diffusion-based recommendation models user preferences and item representations through a progressive denoising process. Unlike autoregressive approaches that generate discrete tokens sequentially, diffusion models learn to recover recommendation-related signals from corrupted observations. For a continuous target $\mathbf{x}_0$, a standard Gaussian forward process [7] is defined as

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}\right), \tag{4}$$

where $\bar{\alpha}_t$ denotes the cumulative noise schedule. A parameterized denoising model learns to reverse this corruption process, optionally conditioned on user interaction histories. Depending on the target representation, the resulting outputs can be reconstructed interaction scores, item embeddings, or preference representations used to derive recommendations.

3.3.1 Interaction and Item Representation Diffusion. Early diffusion-based recommenders explore generative modeling of user–item interactions. DiffRec [37] applies diffusion to interaction vectors and reconstructs personalized preference signals for item ranking. Its forward process introduces controlled noise to preserve informative interaction patterns during denoising. To address the characteristics of recommendation data, DiffRec further introduces two extensions. L-DiffRec compresses item dimensions through clustering and latent-space diffusion, while T-DiffRec incorporates temporal information through interaction reweighting. These designs adapt diffusion to large item spaces and evolving user preferences.

Another research direction generates representations of target items conditioned on historical interactions. DreamRec [40] reformulates sequential recommendation as conditional item generation. A Transformer encodes the user's interaction history into a guidance representation, which conditions the reverse diffusion process to generate an embedding corresponding to a preferred next item. The generated representation is subsequently matched with catalog items through embedding similarity. This formulation shifts the prediction target from candidate relevance scores to a distribution over desirable item representations.

DiffuRec [19] similarly employs diffusion for sequential recommendation, but emphasizes uncertainty in item representations. It models target items through distributions that capture multiple item aspects and user interests. Historical interactions guide the

reconstruction of target item representations, followed by a rounding operation that identifies discrete items. Together, DreamRec and DiffuRec demonstrate how conditional diffusion can represent personalized preferences beyond fixed item embeddings.

DimeRec [18] further investigates the interaction between generative modeling and nonstationary user interests. It combines a guidance extraction module with a diffusion-based aggregation module. The former extracts stable guidance signals from evolving interaction histories, while the latter generates preference representations conditioned on these signals. This design integrates multi-interest modeling with diffusion-based representation generation.

These approaches differ primarily in their generation targets and conditioning mechanisms. DiffRec reconstructs interaction-level preferences, whereas DreamRec and DiffuRec generate representations associated with target items. DimeRec emphasizes the aggregation of dynamic user interests. The choice of target space determines how denoised outputs are converted into item rankings or recommendations.

3.3.2 Discrete-State and Catalog-Aware Generation. Continuous diffusion models operate in learned representation spaces, while recommendation decisions ultimately concern discrete catalog items. Recent methods investigate this connection through explicit item mapping and discrete-state diffusion.

DCRec [9] introduces a discrete-to-continuous diffusion framework for sequential recommendation. It extends the generative process to model the transition between continuous item representations and discrete item indices. Its dual conditional diffusion Transformer incorporates implicit and explicit conditioning to capture historical preferences. This formulation integrates item-level prediction more closely with the diffusion process.

DDSR [39] explores an alternative approach by modeling diffusion directly in a discrete state space. Motivated by uncertainty in sequential user behavior, DDSR represents interaction sequences through fuzzy sets and employs discrete transition processes to model evolving interests. To manage the large state space associated with item identities, it introduces compact semantic labels obtained through quantization.

These methods extend diffusion-based recommendation beyond conventional continuous embedding generation. DCRec connects continuous denoising with discrete item prediction, whereas DDSR models transitions over discrete semantic states. Their designs highlight the importance of representation granularity and state-space construction in connecting generative processes with catalog-level recommendation decisions.

3.3.3 Preference Alignment and Efficient Inference. A central challenge in diffusion-based recommendation is the relationship between denoising objectives and personalized ranking quality. A common noise-prediction objective minimizes

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}} \left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, h)\|_2^2 \right], \tag{5}$$

where $\boldsymbol{\epsilon}$ denotes sampled Gaussian noise and h represents the interaction history. Although denoising objectives support distribution learning, reconstruction accuracy and recommendation relevance capture different aspects of model performance.

PreferDiff [26] addresses this relationship by incorporating personalized ranking into diffusion learning. It reformulates Bayesian personalized ranking within a generative likelihood framework and introduces multiple negative samples to capture relative item preferences. Variational optimization and cosine-based error further connect diffusion learning with ranking objectives. This approach integrates preference-aware supervision into the generative training process.

Diffusion-generated preferences can also support existing sequential recommenders. PDRec [28] employs a diffusion model as a plug-in component to estimate dynamic user preferences. It uses these signals for historical behavior reweighting, positive-sample augmentation, and negative-sample selection. This design uses diffusion-based preference modeling to enhance downstream recommendation without directly generating the final item.

Inference efficiency constitutes another important research direction. Conventional diffusion sampling often involves multiple denoising steps, creating computational overhead in latency-sensitive recommendation settings. TA-Rec [29] addresses this trade-off through a two-stage framework for one-step generation. Temporal consistency regularization is employed during pretraining to encourage consistency across denoising steps, while adaptive preference alignment during fine-tuning improves the correspondence between generated representations and user preferences.

These methods address complementary aspects of diffusion-based recommendation. PreferDiff focuses on aligning generative learning with ranking objectives, PDRec integrates generated preference signals into sequential recommendation, and TA-Rec improves the efficiency of the generation process. Their contributions connect diffusion modeling with recommendation utility and practical inference requirements.

Overall, diffusion-based recommendation provides a distributional modeling framework for generating interaction preferences, target item representations, and discrete recommendation states. The development of this paradigm has progressed from preference reconstruction toward conditional generation, structured item modeling, and preference-aware optimization. Its effectiveness depends on the interaction among generation targets, conditioning information, learning objectives, and inference efficiency.

4 Cross-Paradigm Comparison

The three generative paradigms differ in their output representations, learning objectives, and mechanisms for producing catalog-level recommendations. These distinctions influence both the statistical properties of the models and their computational requirements. The following comparison examines their generation targets, learning signals, inference procedures, and principal trade-offs.

4.1 Representation and Inductive Bias

The representation of recommendation targets determines the structure of the prediction space and the information shared across items. Natural-language representations leverage semantic knowledge acquired during pretraining, allowing models to associate user preferences with item descriptions and attributes. Their effectiveness depends on the correspondence between textual expressions and catalog identities, particularly when item names are ambiguous or descriptions are incomplete.

Semantic identifiers provide a more structured alternative. By representing items as sequences of discrete codes, methods such as TIGER [33] and LC-Rec [44] replace large atomic item vocabularies with compositional token representations. Shared code prefixes enable parameter sharing across related items, while the learned code structure introduces inductive biases into the generative prediction process. Collaborative and multimodal tokenization further incorporate behavioral relationships that may differ from content-based similarity.

Continuous representations offer another modeling perspective. Diffusion-based approaches generate interaction preferences or target item embeddings within continuous spaces, where similarity is determined by the learned representation geometry. DiffRec [37] reconstructs interaction-level signals, whereas DreamRec [40] generates embeddings associated with preferred items. The quality of the resulting recommendations depends on both the generated representation and its correspondence to catalog items.

These representation choices establish different forms of generalization. Text supports transfer from linguistic knowledge, semantic codes encourage information sharing through structured identifiers, and continuous embeddings support distributional modeling in latent spaces. Their respective advantages depend on the availability of semantic features, collaborative signals, and the structure of user preferences.

4.2 Learning Objectives and Decision Alignment

The learning objectives of generative recommenders reflect different assumptions about how user preferences should be modeled. Autoregressive approaches typically maximize the conditional likelihood of target item sequences. For semantic-ID generation, this objective decomposes item prediction into successive token-level decisions, allowing statistical dependencies to be learned across identifier components.

Diffusion-based approaches instead optimize reconstruction or denoising objectives. These objectives encourage models to recover interaction patterns or target representations from corrupted observations. However, representation reconstruction and recommendation ranking optimize different quantities. In particular, small errors in an embedding space may change the nearest catalog item, while accurate reconstruction of historical interactions may not directly improve future recommendation relevance.

Preference-oriented training introduces additional supervision to connect generative learning with recommendation utility. PreferDiff [26] incorporates pairwise preference modeling into diffusion learning, while OneRec [5] uses iterative preference alignment to optimize generated recommendation sessions. These approaches extend the training signal beyond the likelihood or reconstruction of observed behavior toward relative preference estimation.

A common consideration across these objectives is the relationship between observed interactions and underlying user preferences. Historical feedback reflects both user interests and the exposure policies under which items were presented. Consequently, likelihood estimation, denoising reconstruction, and preference alignment capture different aspects of the recommendation problem. Their effectiveness is ultimately determined by the correspondence between training objectives and the quality of the resulting recommendation decisions.

4.3 Inference, Validity, and Catalog Dynamics

The output representation also determines the operations required during inference. Language-based methods generate textual descriptions or queries that must be resolved through matching or retrieval. Semantic-ID models generate structured token sequences and use identifier lookup to recover catalog items. Diffusion models typically require denoising followed by preference-score ranking or similarity-based retrieval, while discrete-state diffusion employs transitions over learned semantic states.

These inference procedures introduce distinct computational characteristics. Autoregressive decoding cost depends on sequence length and search strategy, with beam search increasing candidate exploration at additional computational cost. Diffusion inference depends on the number of denoising steps and the complexity of the generated representation. Models such as TA-Rec [29] reduce this cost through one-step generation. Sequential architectures such as HSTU [41] instead emphasize efficient computation over large-scale interaction histories.

The generated outputs must ultimately satisfy catalog-level constraints. Textual outputs require entity resolution, semantic codes require valid identifier mappings, and continuous embeddings require a decision rule for selecting discrete items. Catalog filtering, deduplication, and candidate ranking may therefore remain part of the serving pipeline regardless of the underlying generative paradigm.

Catalog evolution further connects representation design with system maintenance. New items can be incorporated into embedding indexes when suitable representations are available, whereas semantic-ID systems additionally require code assignment and identifier management. Changes to learned codebooks may affect previously established item mappings. Text-based approaches can exploit newly available descriptions, while their retrieval quality depends on the corresponding catalog index.

The practical efficiency of generative recommendation should therefore be evaluated at the system level. Relevant costs include model execution, decoding or denoising, catalog mapping, filtering, and any subsequent ranking operations. This perspective provides a common basis for comparing architectures with substantially different inference procedures.

4.4 Evaluation and Cross-Paradigm Comparability

A meaningful comparison across generative paradigms requires consistent recommendation tasks and evaluation settings. Models may produce the same number of final recommendations while performing substantially different amounts of intermediate computation. For example, a semantic-ID generator may explore multiple decoding paths, whereas a diffusion model may generate a continuous representation and retrieve its nearest catalog items. Comparing their final top-K results alone does not capture these differences.

Evaluation should therefore distinguish recommendation effectiveness from inference efficiency. Ranking metrics such as Recall@K and NDCG@K measure the relevance of final recommendation lists, while latency, throughput, and memory consumption characterize serving requirements. Reporting recommendation quality under comparable computational budgets offers further insight into the trade-offs among generative paradigms.

The candidate universe and evaluation protocol also influence reported performance. Candidate-based prediction evaluates selection within a restricted set, whereas full-catalog retrieval requires identifying relevant items among all eligible catalog entries. Comparisons should use consistent temporal splits, interaction histories, catalog snapshots, and candidate eligibility rules. When sampled candidates are used for evaluation, the sampling procedure and candidate-set size should be reported explicitly.

Representation construction introduces further considerations for temporal evaluation. Semantic tokenizers and item embedding models may use information that becomes available at different stages of the recommendation process. Separating training-period interactions from future behavioral signals is necessary to assess generalization under realistic catalog conditions. Performance on new items and long-tail items can additionally reveal differences that aggregate ranking metrics may obscure.

Finally, comparisons should account for the complete recommendation pipeline, including interactions among its components. Generative retrieval, catalog grounding, reranking, and preference alignment may be optimized separately or jointly, leading to different degrees of system integration. Evaluating final recommendation quality alongside intermediate validity, decoding cost, and serving efficiency provides a more comprehensive characterization of each approach.

Overall, the three paradigms make distinct design choices and use generation mechanisms that are not directly interchangeable. Language-based methods emphasize semantic flexibility. Recommendation-specific generative systems exploit structured item representations and sequential prediction. Diffusion-based approaches model preferences through denoising and distributional generation. Their relative effectiveness depends on how representation learning, optimization objectives, inference mechanisms, and catalog decisions are coordinated within the complete recommendation system.

5 Open Challenges and Research Directions

Despite recent advances, generative recommendation faces several challenges in representation learning, preference modeling, inference efficiency, and practical deployment. We highlight six directions for future research.

Adaptive representations and catalog evolution. Dynamic catalogs require stable item representations while accommodating newly introduced items. Semantic-ID methods face code collisions, uneven code utilization, and identifier changes. Future research should explore incremental tokenization, stable codebooks, and the integration of semantic and collaborative information to improve cold-start and long-tail recommendation.

Valid and controllable generation. Generated outputs must correspond to available items and satisfy user and application constraints. Promising directions include catalog-aware decoding, constraint-aware generation, and list-level optimization. These mechanisms should jointly consider recommendation relevance, validity, diversity, and catalog coverage.

Preference alignment and feedback modeling. Generative objectives based on interaction likelihood or denoising reconstruction may differ from actual recommendation utility. Future work could investigate preference-aware learning, uncertainty-aware reward modeling, and multi-objective optimization to better align generated recommendations with user preferences and long-term outcomes.

Efficient inference. Autoregressive decoding and iterative denoising introduce computational overhead in large-scale recommendation. Adaptive decoding, model distillation, and efficient sampling offer promising directions for reducing latency. Joint optimization of generation and catalog retrieval may further improve end-to-end efficiency.

Reliable evaluation and reproducibility. Fair comparison requires consistent temporal splits, candidate sets, and computational budgets. Model-scaling studies [23] and analyses of sampled ranking metrics [14] highlight important evaluation considerations. Future benchmarks should incorporate cold-start performance, exposure bias, generation validity, and computational efficiency. Language-based methods additionally require evaluation of history sensitivity and output reliability [11].

Privacy and trustworthy deployment. Rich interaction histories introduce risks of sensitive information exposure and model memorization. Privacy-preserving learning, access control, and reliable output filtering are important directions for developing trustworthy generative recommenders.

6 Conclusion

This survey systematically reviewed three major paradigms of generative recommendation: language-model-based approaches, recommendation-specific generative systems, and diffusion-based methods. Through a comparative analysis of their representations, learning objectives, and inference mechanisms, we identified the distinct modeling characteristics and practical trade-offs of each paradigm. A central finding is that recommendation effectiveness depends not only on generative modeling capabilities but also on how generated outputs are mapped to catalog items and aligned with user preferences. Future research should emphasize adaptive representations, preference-aware optimization, and efficient generation, supported by rigorous evaluation under realistic recommendation settings.

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